

Generation Interconnection System Impact Study Report

For Queue Project Q17-0840

Pine Hollow 138 kV

December 04, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0840. The Interconnection Customer proposes a 289.7 MW natural gas combined cycle facility interconnecting at the Pine Hollow 138 kV substation in the Virginia area. The request was received on August 03, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 340 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,254,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Gordonsville to Proffit 138.0 kV L
Contingency: Fluvanna P.S to Louisa Generation Facility 345.0 kV L
Loading: 154.0% (Rating: 191 MVA)
DFAX: 35.3%
Pre-project loading: 100.4%
2. Facility: Colgate to Proffit 138.0 kV Line
Contingency: Louisa Generation Facility to Fluvanna P.S 345.0 kV
Loading: 153.7% (Rating: 202 MVA)
DFAX: 35.3%
Pre-project loading: 103.1%
3. Facility: Albreda Tans Mtn Tap to Colgate 138.0 kV line
Contingency: Fluvanna P.S to Louisa Generation Facility 345.0 kV Line
Loading: 151.6% (Rating: 228 MVA)
DFAX: 35.3%
Pre-project loading: 106.8%
4. Facility: Charlottesville to Pine Hollow 138.0 kV
Contingency: Albreda Tans Mtn Tap to Colgate 138.0 kV
Loading: 138.7% (Rating: 243 MVA)
DFAX: 100.0%
Pre-project loading: 19.6%
5. Facility: Brema Bluff to Amber 138.0 kV Line
Contingency: Bluff City to Lake Lure 138.0 kV Line
Loading: 127.5% (Rating: 194 MVA)
DFAX: 8.3%
Pre-project loading: 115.1%
6. Facility: Peninsula to Pine Hollow 138.0 kV
Contingency: Louisa Generation Facility to Fluvanna P.S 345.0 kV line
Loading: 121.6% (Rating: 240 MVA)

DFAX: 35.3%
Pre-project loading: 79.0%

7. Facility: Peninsula to Running Y 138.0 kV Line
Contingency: Louisa Generation Facility to Fluvanna P.S 345.0 kV
Loading: 121.1% (Rating: 235 MVA)
DFAX: 35.3%
Pre-project loading: 77.6%

8. Facility: Running Y to Albreda Tans Mtn Tap 138.0 kV
Contingency: Fluvanna P.S to Louisa Generation Facility 345.0 kV
Loading: 119.3% (Rating: 231 MVA)
DFAX: 35.3%
Pre-project loading: 75.2%

9. Facility: Fluvanna P.S to Grove Street 138.0 kV
Contingency: Albreda Tans Mtn Tap to Colgate 138.0 kV
Loading: 104.1% (Rating: 345 MVA)
DFAX: 39.7%
Pre-project loading: 70.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gordonsville to Proffit 138.0 kV L	Rebuild line with higher-capacity conductor	\$11,385,000	61% (\$6,915,000)
Colgate to Proffit 138.0 kV Line	Rebuild line with higher-capacity conductor	\$6,539,000	61% (\$3,972,000)
Albreda Tans Mtn Tap to Colgate 138.0 kV	Rebuild line with higher-capacity conductor	\$2,784,000	61% (\$1,691,000)
Charlottesville to Pine Hollow 138.0 kV	Rebuild line with higher-capacity conductor	\$9,151,000	80% (\$7,275,000)
Bremo Bluff to Amber 138.0 kV Line	Rebuild line with higher-capacity conductor	\$14,878,000	33% (\$4,976,000)
Peninsula to Pine Hollow 138.0 kV	Rebuild line with higher-capacity conductor	\$6,073,000	65% (\$3,934,000)
Peninsula to Running Y 138.0 kV Line	Rebuild line with higher-capacity conductor	\$7,491,000	65% (\$4,852,000)
Running Y to Albreda Tans Mtn Tap 138.0 kV L	Rebuild line with higher-capacity conductor	\$6,605,000	65% (\$4,278,000)
Fluvanna P.S to Grove St 138.0 kV	Reconductor line	\$562,000	64% (\$361,000)
	Total allocated to project		\$38,254,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.